

Breathing In, Breathing Out — Paper B

MEDIUM • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- A calculator is allowed, but every number here is designed to work without one.
- Where a question asks about air moving in or out, your answer must follow the chain in order: **muscles change the volume, the volume change causes a pressure change, air then flows down the pressure gradient**. A muscle never sucks or pushes air by itself.
- The mark for each question is shown in square brackets on the right.
- You may need: **$1000\text{ cm}^3 = 1\text{ dm}^3$** .

SECTION 1 – WHAT GOES IN IS NOT WHAT COMES OUT

1. The table shows the composition of inspired and expired air.

Gas	Inspired	Expired
Oxygen	21%	16%
Carbon dioxide	0.04%	4%
Nitrogen	78%	78%

- (a) Name the gas whose percentage does not change, and explain why it does not. [2]

- (b) Calculate how many times more carbon dioxide there is in expired air than in inspired air. [1]

- (c) State by how many percentage points the oxygen falls. [1]

- (d) Name one way, not shown in the table, in which expired air differs from inspired air. [1]

2. A student blows through a tube into a flask of limewater. A second flask of limewater has ordinary room air drawn through it at the same rate.

(a) State what happens in the flask the student blows into. [1]

(b) The second flask changes too, but takes far longer. Explain why. [2]

SECTION 2 – THE AIRWAYS

3. The trachea and bronchi are lined with goblet cells and ciliated cells. Describe how these two cell types work together to keep the lungs clean. [3]

4. The trachea is held open by C-shaped rings of cartilage. Explain what would happen when you breathed in hard if these rings were not there. [2]

5. Give two reasons why breathing in through the nose is better for the lungs than breathing in through the mouth. [2]

SECTION 3 – BREATHING AND EXERCISE

6. At rest, Nikhil takes 12 breaths each minute and moves 500 cm^3 of air with every breath. While he is running he takes 24 breaths each minute and moves 2000 cm^3 with every breath.

- (a)** Calculate the volume of air he breathes each minute at rest. Give your answer in dm^3 . [2]

- (b)** Calculate the volume of air he breathes each minute while running, in dm^3 . [2]

- (c)** State how many times greater the running value is, and name the two things about his breathing that changed. [2]

Work in cm^3 throughout, then convert once at the very end. Mixing the two units halfway is the usual way marks are lost here.

- 7.** Explain, step by step, why breathing becomes faster and deeper during exercise. Begin with what the muscles are doing. [4]

- 8.** A student wants to measure her resting breathing rate accurately. Describe how she should do it, and give two things she must keep the same each time she repeats the measurement. [3]

- 9.** Explain why expired air is warmer and carries more water vapour than the air that [2]
was breathed in.

End of paper. Check every answer has a unit or a form the question actually asked for.